Abdalrahman Mohammed Tawfik

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About me

Graduated Engineer from the Higher Technological Institute with a degree in Communication and Electronics Engineering. I have a strong foundation in analog, digital, and mobile communication systems, along with essential knowledge of IP networking and mobile technologies. My graduation project focused on enhancing BER and PAPR using FBMC and UFMC modulation techniques.

Education

Higher Technological Institute — 10th of Ramadan City, Egypt
Bachelor of Engineering, Communication and Electronics Department — *Oct. 2020 – May. 2025*

- Relevant Coursework: Analog and Digital Communication, Antenna, Electronic Circuits — GPA: 2.78
- Graduation Project: **Enhancing BER and PAPR using FBMC and UFMC**

Training

CCNP Security SCOR (350-701) — May. 2024 – Aug. 2024
T-Shoot Company — *Onsite*

- Focused on enterprise security, VPNs, network access control, and threat defense techniques.

CCNA (200-301) — May. 2023 – Aug. 2023
T-Shoot Company — *Onsite*

- Covered core networking skills: IP addressing, routing switching, network protocols, and troubleshooting.

Industrial Training — May. 2022 – Aug. 2022
Oulabitex Textile Company — *Onsite*

- Gained exposure to classic control systems and industrial components used in textile machinery. Observed machine operation and control panel basics.

Projects

Enhancing BER and PAPR Using FBMC and UFMC | *MATLAB* — Aug. 2023 – Jan. 2024

- Investigated advanced multicarrier modulation techniques — **FBMC** and **UFMC** — to optimize Bit Error Rate and Peak-to-Average Power Ratio in next-generation communication systems.
- Simulated BER and PAPR performance in MATLAB and evaluated their efficiency compared to OFDM in 5G use cases.

Simulation of Multipath Fading over Rayleigh Channel | *Simulink – MATLAB* — Mar. 2024

- Simulated wireless channel behavior using a **Rayleigh fading model** to analyze multipath effects in mobile communications.
- Tested modulation schemes (BPSK, QPSK, 16-QAM) to study Bit Error Rate and data throughput under various fading conditions.

Linear Block Code Circuit for Error Detection & Correction | *Proteus* — Jan. 2024

- Designed and implemented a digital circuit using **Hamming codes** to perform error detection and correction in transmitted binary data.
- Applied concepts of syndrome decoding and parity checks for channel coding, aligning with real-world wireless communication standards.

Skills

Programming Languages: Python, MATLAB

Simulation and Design Tools: CST Studio Suite, Proteus, MATLAB, Simulink

Networking Tools: Wireshark, GNS3, VMware, Packet Tracer

Software and IDEs: Visual Studio, MS Office

Soft Skills: Problem-solving, Teamwork, Fast learner, Adaptability, Time management

Languages: Arabic (Native), English (Intermediate to Upper-Intermediate – B1–B2)

Extracurricular Activities

Member, Computer Science Committee – HTI Student Branch Sep. 2023 – Present

- Participated in technical learning sessions focused on **C++ programming** and **problem-solving fundamentals**.
- Completed basic tasks and exercises as part of the committee's internal development phase.